

EE 556: Nuclear Reactor Control

Course Code	EE 556
Course Name	Nuclear Reactor Control
L-T-P-C	1-0-0-1
Prerequisites	Differential Equations, Laplace Transform, Elementary Matrix Algebra
Intended for	UG (3rd year and final year) and PG students
Mutual Exclusion	None
Approval	54th BoA (May 2024)

Course Contents

- Introduction: India's nuclear energy scenario, Nuclear Power Plants (NPP) Schematic, Components of a NPP, Nuclear Reactor types and applications. (1 Hour)
- Selected Topics from Reactor Physics: Interactions of neutron with nuclei (matter), Nuclear Reaction Cross—sections, Mechanism of nuclear fission, Nuclear fission reactions, Energy released in fission; Prompt and delayed neutrons, Multiplication factor, Four factor formula, Non-leakage probability; The one – speed neutron diffusion equation, General solution of one – speed neutron diffusion equation; Reactor criticality condition for parallelepiped and cylindrical geometries, Effect of reflector. (3 Hours)
- Nuclear Reactor Kinetics: Derivation of point kinetics model from one speed neutron diffusion equation; Solution of point kinetics model for step change in reactivity, reactor period, In-hour equation, Interpretation of reactor shut-down and trip. (2 Hours)
- Reactor as a Control Element: Linearization of point kinetics model and representation into standard state-space form, transfer function, Reactor stability, Controllability and Observability. (2 Hours)
- Reactor Dynamics Studies: Modeling of Internal feedback effects due to changes in fuel, coolant and Moderator temperatures; Modeling of Internal feedback effect due to fission product xenon; xenon spatial instability; Modeling of Internal feedback effect due to fission product samarium; Issues in modeling and control of large reactors. (3 Hours)
- Reactor Instrumentation: Methods of neutron flux measurement; n , $\log n$, dn/dt and $d(\log n)/dt$ signals for reactor control, Thermal power measurement; Power correction; Flux mapping. (1 Hour)
- Miscellaneous Topics: Examples of typical reactor power control systems; Reactivity estimation – Inverse point kinetics method, Kalman Filter method. (2 Hours)

Laboratory Modules

- Not envisaged

Text Books

1. James J. Duderstadt and Louis J. Hamilton, **Nuclear Reactor Analysis**, Wiley, 1976
2. L. E. Weaver, **Reactor Dynamics and Control: State Space Techniques**, American Elsevier Publishing Company, 1968

References

1. A. P. Tiwari, et. al., **Modelling and Spatial Control of 540 MWe Pressurized Heavy Water Reactor**, Trans. INAE, Vol. 6, pp. 731–753, Sept. 2021.
2. P.V. Surjagade, et. al., **Robust Optimal Integral Sliding Mode Controller for Total Power Control of Large PHWRs**, IEEE Trans. Nucl. Sci., Vol. 65, Issue 7, pp. 1331-1344, 2018.
3. C. S. Subudhi, et. al., **A mathematical model for total power control loop of large PHWRs**, IEEE Trans. Nucl. Sci., vol. 63, no. 3, pp. 1901–1911, June 2016.
4. Patel SB, Mukhopadhyay S, Tiwari A, **Estimation of reactivity and delayed neutron precursors' concentrations using a multi-scale extended kalman filter**, Ann Nucl Energy 111:666–675, 2018.
5. Bhatt TU, Patel SB, Tiwari AP, **Reactivity and delayed neutron precursors' concentration estimation based on recursive nonlinear dynamic data reconciliation technique**, IEEE Trans Nucl Sci 66(2):541–548, 2019.
6. Mishra AK, Shimjith SR, Bhatt TU, Tiwari AP, **Kalman filter-based dynamic compensator for vanadium self powered neutron detectors**, IEEE Trans Nucl Sci 61(3):1360–1368, 2014.
7. M. G. Na, I. J. Hwang, and Y. J. Lee, **Design of a fuzzy model predictive power controller for pressurized water reactors**, IEEE Transactions on Nuclear Science, vol. 53, no. 3, pp. 1504–1514, June 2006.
8. R. N. Banavar and U. V. Deshpande, **Robust controller design for a nuclear power plant using h-infinity optimization**, IEEE Transactions on Nuclear Science, vol. 45, no. 2, pp. 129–140, Apr 1998.
9. H. Eliasi, M. Menhaj, and H. Davilu, **Robust nonlinear model predictive control for a PWR nuclear power plant**, Progress in Nuclear Energy, vol. 54, no. 1, pp. 177 – 185, 2012.